

DIOGO SANTOS
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ABOUT ME

I am a Jack-of-all-trades, keen on mastering them all. One of my main motivations is to continuously add value to the roles I perform, finding ways to optimize my work by applying my talents and capabilities, even in contexts that may seem unrelated, at first sight. I have a particular interest in areas such as Machine Learning, Computer Vision, and Robotics, as well as cybersecurity, data privacy and software resilience. Outside computers, I am a poetry writer, music lover, avid crocheter and a decent juggler.

EDUCATION

Integrated Master's in Informatics and Computing Engineering 2011 - 2016

University of Porto - Faculty of Engineering (BSc and MSc degree)
Final grade: 18/20 - Grade A on the European grading scale

Secondary Education in Sciences and Technologies 2008 - 2011

Escola Secundária Garcia de Orta
Final grade: 18/20

WORK EXPERIENCE

IT Officer - Software Development and Data Engineering Frankfurt, Germany *AML - EU Anti-Money Laundering Authority* May 2026-Present

- Leading the design of a secure case collaboration platform with Double Key Encryption.
- Automating case provisioning and access management with Power Automate.
- Authoring the organization's Secure Software Development Lifecycle (SSDLC) strategy.
- Defining the CI/CD and vulnerability management process for the EuReCA project.
- Administering GitHub Enterprise and developing supporting Azure services.
- Translating business and compliance requirements into technical designs.

Backend Team Coordinator and Software Engineer Braga, Portugal *PeekMed - PeekHealth, S.A.* Nov 2022-Apr 2026

- Led the backend team from March 2025.
- Bridged product, QA, AI and leadership teams in a QMS/ISMS-regulated environment.
- Developed secure backend services and APIs with Node.js, TypeScript and MongoDB.
- Ensured code quality and security through code reviews, testing and threat modeling.
- Developed the C++/Qt desktop application for AI-based orthopedic surgery planning.
- Integrated 3D medical imaging (VTK, ITK) and DICOM/PACS and led R&D initiatives.

Software Developer Porto, Portugal *Freelance* Sep 2023-Mar 2026

- Developed software and documentation for clients such as the Municipality of Porto.
- Developed C-ITS software following European standards and data models, such as DATEX-II.
- Developed CCTV and video analytics systems for road safety, feeding machine learning data flows.

Software Engineer in the field of C-ITS Porto, Portugal *Municipality of Porto - Câmara Municipal do Porto* Apr 2020-Nov 2022

- Integrated an innovative C-ITS software engineering team as lead engineer,

prototyping, developing and documenting C-ITS software.

- Developed public tender specifications and applications, and managed public projects, including projects funded by the European Union.
- Participated in national and European technical harmonization and standardization activities in the field of C-ITS (C-Roads European Platform and C-Roads Portugal)
- Performed data safety analysis and GDPR compliance activities.
- Developed and maintained internal management and open data platforms and APIs.

Control Software Engineer

ESO - European Southern Observatory

Munich, Germany

Jul 2017-Jul 2019

- Developed control software for the European Extremely Large Telescope (E-ELT).
- Prototyped, developed, and documented software under strict requirements.
- Implemented low-level, high-performance, control loops using C and C++.
- Implemented high-level tools and interfaces using Python and Qt.
- Developed and tested automated deployment solutions and DevOps prototypes.

Researcher and developer

LIACC - Artificial Intelligence and Computer Science Laboratory

Porto, Portugal

Jul 2016-Apr 2017

- Applied Computer Vision methodologies to *Intelligent Transportation Systems*.
- Integrated research to industry settings, collaborating with the company *Armis Group*.

Systems integrator and developer

Multi-agent System for Disruption Management - MASDIMA, Lda.

Porto, Portugal

Oct 2014-Sep 2015

- Developed and maintained multi-agent protocols and algorithms for airline disruption management.
- Integrated and documented pre-existent systems.

TECHNICAL SKILLS

- **Operating Systems:** Windows (4/5), Linux (3/5), Android (3/5), Mac OS (2/5), iOS (1/5)
- **Programming Languages:** C++ (4/5), Java (4/5), Python (4/5), TypeScript (4/5), JavaScript (4/5), PHP (3/5), C# (3/5), C (3/5), PowerShell (3/5)
- **Web Technologies:** HTML + CSS (4/5), Node.js (4/5), Spring Framework (3/5), AngularJS (2/5), ASP.NET (1/5)
- **Databases:** PostgreSQL (4/5), SQLite (4/5), MySQL / MariaDB (4/5), MongoDB (4/5), SQL Server (3/5), Redis (2/5)
- **Cloud Platforms and Services:** Microsoft 365 / SharePoint (4/5), Power Automate (4/5), Microsoft Azure (3/5), AWS (2/5)
- **Machine Learning & Computer Vision Tools:** RapidMiner (3/5), OpenCV (3/5), TensorFlow (2/5)
- **Continuous Integration and Deployment Tools:** GitHub Actions (3/5), Jenkins (3/5), HashiCorp Nomad (3/5), HashiCorp Terraform (3/5)
- **Testing Tools:** Jest (4/5), Cypress (3/5)
- **Other Tools and Frameworks:** Qt (4/5), LaTeX (4/5), Docker (3/5), FFmpeg (3/5), Grafana (3/5)

PROFESSIONAL DEVELOPMENT

- **The Complete Guide to Becoming a Software Architect** · Udemy Jul 2023
- **Design Patterns in Modern C++** · Udemy Dec 2022
- **Qt 5 Core Advanced with C++** · Udemy Nov-Dec 2022
- **Data Scientist Fundamentals - training course** · INA Apr-May 2022
- **General Cybersecurity Course - A whole of society perspective** · CNCS Dec 2021

PUBLICATIONS

- **The ELT M1 Local Control Software: From Requirements to Implementation** 2019
2019 17th Biennial International Conference on Accelerator and Large Experimental Physics Control Systems (ICALEPCS2019)
Authors: L. Andolfato, J. Argomedo, C. Diaz Cano, R. Frahm, T. R. Grudzien, N. Kornweibel, D. Ribeiro Gomes Santos, J. Sagatowski, C. M. Silva
- **A Methodological Approach for Inferring Urban Indicators Through Computer Vision** 2018
2018 IEEE International Smart Cities Conference (ISC2)
Authors: Sara Paiva, Diogo Santos, Rosaldo J. F. Rossetti
- **Adding the Third Dimension to Urban Networks for Electric Mobility Simulation: An Example for the City of Porto** 2018
Developments and Advances in Intelligent Systems and Applications
Authors: Diogo Santos, José Pinto, Rosaldo J. F. Rossetti, Eugénio Oliveira
- **Towards the Integration of Electric Buses in Conventional Bus Fleets** 2016
2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)
Authors: Diogo Santos, Zafeiris Kokkinogenis, Jorge Freire de Sousa, Deborah Perrotta, Rosaldo J. F. Rossetti
- **Integrating Electric Buses in Conventional Public Transit: A First Appraisal** 2016
Master's Dissertation in Informatics and Computing Engineering, FEUP
- **A Weighted Constraint Optimization Approach to the Nurse Scheduling Problem** 2015
2015 IEEE 18th International Conference on Computational Science and Engineering (CSE)
Authors: Diogo Santos, Pedro Fernandes, Henrique Lopes Cardoso, Eugénio Oliveira

DISSERTATION SUPERVISION

- **A Data Analysis Pipeline for the Study and Categorization of User Content in Online Health Communities** 2022
MSc Dissertation, FEUP · Author: Sara Silva · Supervisor: Rosaldo J. F. Rossetti
- **Inferring Urban Indicators Through Computer Vision on Google Street View** 2018
MSc Dissertation, FEUP · Author: Sara Paiva · Supervisor: Rosaldo J. F. Rossetti
- **Computer-Vision-based Surveillance of Intelligent Transportation Systems** 2017
MSc Dissertation, FEUP · Author: João Neto · Supervisor: Rosaldo J. F. Rossetti

LANGUAGES

Portuguese	English	Spanish	German
Native language	Advanced (Fluent)	Basic - Intermediary	Basic